

# AI Seasonal Demand Forecasting

Exploratory analysis and model-readiness report

Forecast window: 2026-03-03 to 2026-04-01 | Model: XGBoost daily station demand

| Chronological holdout                    | Training coverage                                               | Forecast coverage                                    |
|------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------|
| MAE 5.15 orders<br>RMSE 8.1<br>R2 0.7806 | 59,690 training rows<br>268 stations<br>Calendar-aware features | 30 days<br>1,004 route allocations<br>4 event alerts |

## What the model uses

Delivered and partially delivered orders are deduplicated from line-item exports before aggregation. The daily model uses station, weekday, month, weekend, known festival/holiday/local-event flags, prior-day demand, a 7-day rolling average, and observed vendor capacity. Features are restricted to information available before the prediction day.

## Operational findings

| Station                          | 30-day orders | Forecast revenue | Readiness | Risk   |
|----------------------------------|---------------|------------------|-----------|--------|
| VIRANGANA LAKSHMIBAI             | 521           | INR 1.64L        | 100/100   | Normal |
| NAGPUR                           | 472           | INR 1.68L        | 100/100   | Normal |
| KANPUR CENTRAL                   | 412           | INR 1.15L        | 100/100   | Normal |
| PT DEEN DAYAL UPADHYAYA JUNCTION | 401           | INR 1.30L        | 100/100   | Normal |
| VIJAYAWADA                       | 375           | INR 1.30L        | 100/100   | Normal |
| NELLORE                          | 345           | INR 1.25L        | 100/100   | Normal |
| BHUSAWAL                         | 341           | INR 1.09L        | 100/100   | Normal |
| WARANGAL                         | 315           | INR 1.17L        | 100/100   | Normal |

## Route and event intelligence

**Highest forecast routes:** Bgkt Mas Sf Exp (20626.0): 95 orders; Mas Bgkt Sf Exp (20625.0): 89 orders; Avadh Exp (19037.0): 58 orders.

**Known calendar alerts:** Holi on 2026-03-04 (1 days into the model window); Eid al-Fitr on 2026-03-19 (16 days into the model window); March local travel peak on 2026-03-20 (17 days into the model window); Ram Navami on 2026-03-26 (23 days into the model window).

## Method and limitations

Route figures are derived by allocating each station's XGBoost forecast using its trailing 90-day train-route demand share. Readiness compares predicted volume with a 10% buffer over recent daily baseline capacity; risk labels are Normal, Medium, High and Critical. These forecasts guide preparation only. Weather, train disruptions, vendor availability, new promotions and local events can change actual demand. Verify local holiday dates and live operations before acting.

Reproduce: activate .venv, run python train\_station\_demand\_forecast.py, then python generate\_eda\_report.py. Supporting notebooks are in notebooks/.